

Ostivarene Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-105/A

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. No changes to the approved analytical procedures are proposed in this submission. Cleaning verification swab results were below the health-based exposure limits derived for the product. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

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P.4 Control of Excipients

For the commercial formulation F-105/A, Aerosil 200 Pharma (Colloidal silicon dioxide, glidant) is used.

Basis for Selection

The basis for selection of Aerosil 200 Pharma is improved flowability during high-speed compression, demonstrated across three pilot-scale batches.

The excipient control strategy was developed in accordance with IPEC Excipient Qualification Guide, ICH Q8(R2) and ICH Q6A.

Registered Specification

Component	Function	Grade per current specification
Colloidal silicon dioxide	glidant	Aerosil 200 Pharma
Reference		SPEC-EX-7468 Rev 09

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Deviations observed during the campaign were investigated and closed with no impact to product quality. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The control strategy links critical quality attributes to critical process parameters identified during development. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.